

Yufei (Haley) Sun

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Education

Johns Hopkins University

MS in Applied Economics — Financial Economics Concentration

Washington, D.C.

Aug 2023

The Ohio State University

B.S. in Financial Mathematics — Actuarial Science Concentration

Columbus, OH

Aug 2020

Technical Skills

Programming Languages: Python (Jupyter), SQL (MySQL, Google BigQuery), R, Stata

Methodologies: Machine Learning, Statistical Inference, A/B Testing, Experimentation, Hypothesis Testing, Probability, Simulations, Confidence Intervals, Correlation, Regression, Time Series, Optimization, Funnel Analysis, Bayesian Inference, Financial Modeling, LLM

Software: Excel (VLOOKUP, Macro, VBA, Pivot), Word, PowerPoint, PowerBI, Tableau, Google Analytics, git, SAP, Salesforce, ERM, CRM

Data Engineering: ETL, Data Extraction, Manipulation and Database Management, AWS, GCP, Snowflake, Databricks, Data Pipelines

Work Experience

NASA, Procurement Department

Data Analyst

Greenbelt, MD

May 2024 – May 2025

- Optimized the enterprise-wide procurement database with SQL to standardize vendor and order data before program launch.
- Built internal and customer-facing dashboards using MicroStrategy to analyze procurement KPIs such as order volume, supplier performance, and cost trends to identify bottlenecks, enhance planning, and increase transparency for government clients.
- Conducted QA and data validation on reports and procurement data to improve accuracy and reliability before release to stakeholders.
- Developed a streamlined product search system that improved accessibility for internal teams and external customers

GTSP Group

Business Analyst Intern

Santa Clara, CA

Nov 2023 – Mar 2024

- Designed, implemented, and tested a customer feedback system to collect user input and improve product functionality and user experience.
- Analyzed holiday promotion performance and campaign profitability using SQL and Excel to provide recommendations that optimized sales strategies and improved ROI.
- Supported development and execution of targeted Amazon ad campaigns, evaluating financial impact and adjusting tactics for positive ROI.
- Built visualized reports with Power BI, presenting key performance insights to executives and supporting data-driven decision-making.

Ernst & Young

Strategy & Analytics Consultant Intern

Shanghai, China

Apr 2021 – Jun 2021

- Performed portfolio analysis for Guotai Junan, a Chinese asset management firm, to consolidate large investment datasets with SQL and Python to evaluate portfolio performance.
- Built valuation models using DCF and Market-based valuation for private equity products and conducted three-statement analysis to measure company growth and profitability
- Collaborated with senior management to design data-driven benchmarking frameworks, enhancing strategic planning and evaluation processes.
- Utilized Bloomberg Terminal to extract and analyze industry and market data to create a performance management dashboard for financial products.

Smart Entry Systems

Marketing Analyst Intern

Miami, FL

Dec 2019 – Feb 2020

- Conducted competitive analysis to analyze consumer behavior and market trends, supporting development of promotional campaigns.
- Built risk models based on historical client behavior to improve product feature design and retention strategies
- Designed personalized client engagement plans by combining market research with data-driven insights to improve customer satisfaction

UnionPay

Procurement Analyst Intern

Shanghai, China

Jun 2018 – Aug 2018

- Created monthly cost and demand reports using Excel which identified sources of excess inventory and reduced carrying costs
- Performed cost-benefit analysis on supplier contracts to create data-driven strategies to improve procurement efficiency.
- Coordinated communications across procurement, finance, and management teams to improve project completion rates

Project Experience

Loan Eligibility Prediction Using Machine Learning Algorithms

Aug 2025 – Nov 2025

- Performed exploratory data analysis and correlation analysis to extract key insights on consumer behavior based on applicant demographic data and conducted data cleaning, including handling missing values and encoding categorical variables.
- Conducted feature engineering to ensure performance, including over-sampling to balance the dataset and scaling numerical variables.
- Created machine learning classification models with various algorithms such as Logistic Regression, K Nearest Neighbor, Decision Tree, and Random Forest to predict loan eligibility.
- Implemented fine-tuning of hyperparameters to enhance model performance and systematically compared models through bootstrapped validation for model selection

Bikes & Cycling Accessories Organization's Transaction Data Base Cohort Analysis

Jun 2023 – Aug 2023

- Developed cohort analysis of bicycle customers to create heatmaps of business KPIs such as monthly retention rate to determine key factors behind customer behavior.
- Carried out RFM analysis to understand purchasing behavior of cohorts to uncover insights into seasonality, promotional effectiveness, and cross-selling opportunities. Investigated factors influencing customer loyalty and engagement with the organization's products.

Mobile Games A/B Testing with Cookie Cats for Player Retention

Mar 2023 – Apr 2023

- Conducted data preprocessing, exploratory data analysis and visualization to extract key insights for player behavior.
- Calculated and compared 1-day retention and 7-day retention for control and treatment group for significant differences.
- Validated the certainty of retention rate differences via bootstrapping to quantify model robustness.